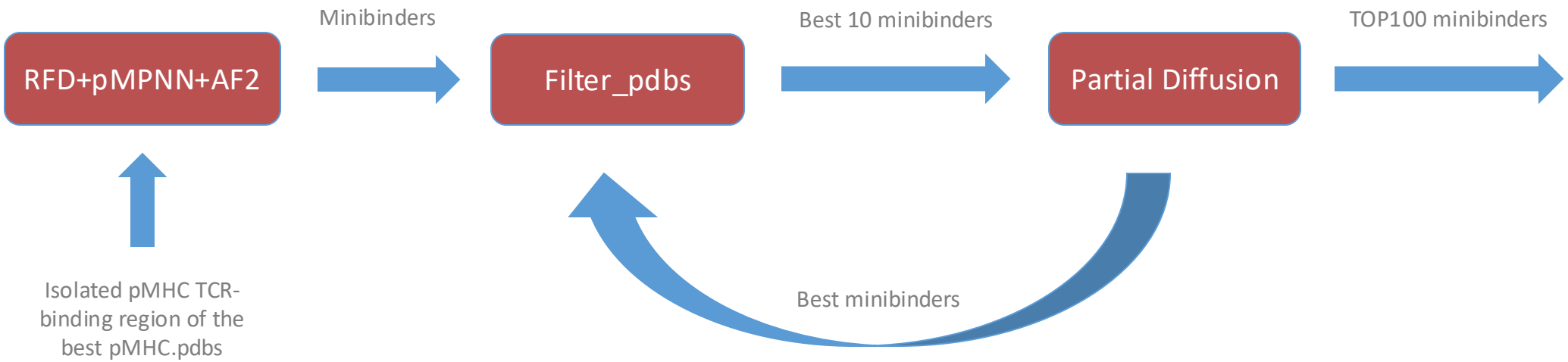


# Serrans RFD

## Versatile Automated Pipeline for De Novo Minibinder Design

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MSc in Pharmaceutical Design and Engineering

# Serrans RFD




Until:

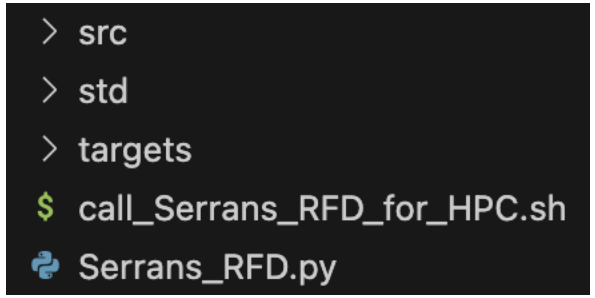
- 100 minibinders with ipAE < 8 and PLDDT > 80
- Max number of PD rounds reached

# Serrans RFD - Arguments

| Argument | Description | Default |
|---|---|---|
| -r | Maximum number of **partial diffusion (PD) rounds** | 3 |
| -m | Minimum number of **minibinders** required | 100 |
| -d | Number of minibinders going to the **next PD round** | 10 |
| -ipae | **ipAE threshold** | 10 |
| -plddt | **pLDDT threshold** | 90 |
| -hs | **Hotsposts** (e.g., A23,A45) – no spaces in between | “False” |
| -sc | Number of **scaffolds** produced by the first RFD round | 175 |
| -seq | Number of **sequences per scaffold** produced in the first RFD round | 5 |
| -psc | Number of **scaffolds** produced in **PD** rounds | 5 |
| -pseq | Number of **sequences per scaffold** produced in the **PD** rounds | 10 |

# Serrans RFD – Execution with Default parameters

1. Upload your target/s pdb files to the **“targets”** folder.
2. Submit **”call_Serrans_RFD_for_HPC.sh”** to the HPC from the “Serrans_RFD” folder.
3. Enjoy! ;)



# Serrans RFD – Execution with your own parameters

1. Upload your target/s pdb files to the **“targets”** folder.
2. Write your arguments on the **”call_Serrans_RFD_for_HPC.sh”** file (line 48).
3. Submit **”call_Serrans_RFD_for_HPC.sh”** to the HPC from the “Serrans_RFD folder”.
4. Enjoy! ;)

# Serrans RFD - Results

1. **“Minibinders_to_validate”** folder contains an Excel file and a ”pdbs” folder.
2. **“TOP10_designs_per_round”** folder contains the TOP10 pdbs that went from one round to the next one.
3. **”Serrans_int_files”** folder contains the “best_designs” and the “Binder_design_outputs” folders from each round.

| pae_interaction | plddt_binder | design_name | sequence |
|---|---|---|---|
| 5,801 | 90,836 | MLEPRTLQY_HLA-A0101_47_dldesign_3_af2pred_110_dldesign_2_af2pred | DEAAARAAQLALIRAIVDRV |
| 5,813 | 93,043 | MLEPRTLQY_HLA-A0101_130_dldesign_4_af2pred_76_dldesign_3_af2pred | EEEAREELRELLRRLRELA |
| 5,889 | 93,099 | MLEPRTLQY_HLA-A0101_48_dldesign_3_af2pred | EEEELEKIKKELKEKLEELL |
| 5,917 | 93,009 | MLEPRTLQY_HLA-A0101_148_dldesign_0_af2pred_139_dldesign_1_af2pred | EEERRRELLELLEELLELAR |